

| Subjects                                        | Subject Type | Activities Planned                                                                                                                                              |                                                                                                                           |                                                          |
|-------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|                                                 |              | Mandatory                                                                                                                                                       | Optional                                                                                                                  | Remarks                                                  |
| Artificial Intelligence on Cloud Computing      | J            | <ul style="list-style-type: none"> <li>• Coursera: AWS: Machine Learning &amp; MLOps Foundations</li> <li>• Internal Hackathon</li> </ul>                       |                                                                                                                           | Both activities are mapped with Assignment 1 and 2 marks |
| Fundamentals of Parallel Processing             | J            | <ul style="list-style-type: none"> <li>• CUDA Toolkit</li> <li>• Participation in SIH</li> </ul>                                                                | Fundamentals of Accelerated Computing with CUDA Python<br>NPTEL: Introduction to parallel programming with OpenMP and MPI | Both activities are mapped with Assignment 1 and 2 marks |
| Web Application Development                     | R            | <ul style="list-style-type: none"> <li>• Capstone Project: Full-stack with AI Bridge</li> <li>• External Hackathon</li> </ul>                                   |                                                                                                                           | Both activities are mapped with Assignment 1 and 2 marks |
| Big Data Analytics                              | J            | <ul style="list-style-type: none"> <li>• Coursera: Real-Time, Real Fast: Kafka &amp; Spark for Data Engineers Specialization</li> <li>• Mini Project</li> </ul> |                                                                                                                           | Both activities are mapped with Assignment 1 and 2 marks |
| Time Series Forecasting and Predictive Analysis | J            | <ul style="list-style-type: none"> <li>• Coursera: Microsoft Power BI Data Analyst Professional Certificate</li> <li>• Internal Hackathon</li> </ul>            |                                                                                                                           | Both activities are mapped with Assignment 1 and 2 marks |
| Data Warehousing and Data Mining                | R            | <ul style="list-style-type: none"> <li>• Case Study: Designing a Data warehouse solution for real-time application</li> <li>• External Hackathon</li> </ul>     | Microsoft Azure Data Fundamentals (DP-900)                                                                                | Both activities are mapped with Assignment 1 and 2 marks |